Janhavi Pohnerkar

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EDUCATION

B-Tech, Electronics and Telecommunication — **2023- Present**
MIT Academy of Engineering
CGPA- 8.41/10

10th | CBSE | Percentage – 89.2% — **2011- 2021**
PSBA English School, Aurangabad

12th | HSC | Percentage – 67.3%
Vidyadham Junior College, Aurangabad — **2021- 2023**

INTERNSHIP EXPERIENCE

Research Intern | 3D Computer Vision — **May 2026- Present**
Indian Institute of Technology Jodhpur (IIT Jodhpur) — ***Jodhpur***

- Exploring diverse areas of 3D computer vision, including CNNs, Vision Transformers, point clouds, mesh and voxel representations, and visual SLAM through research-driven projects.
- Contributed to 360° image stitching and panorama generation for moving vehicles, and implemented and optimized ORB-SLAM3 on embedded and GPU platforms for real-time visual localization.

R&D Intern | Computer Vision | IOT — **June 2025- June 2026**
Copper Cloud Industry 4.0 Solutions — ***Pune***

- Worked on two computer vision and image processing projects focused on quality analysis in the textile and mesh manufacturing industry, from research to deployment.
- Received comprehensive training on IoT architectures and communication protocols, with emphasis on integration and optimization of connected industrial systems for Industry 4.0 applications.

Projects

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- **ORB-SLAM3 Implementation and Optimization on Edge Platforms (IIT Jodhpur)**
  Implemented and optimized the **ORB-SLAM3 visual SLAM** framework on edge platforms (Jetson Orin Nano, Raspberry Pi 4B) for stereo and stereo-inertial configurations, enabling real-time localization and mapping on resource-constrained embedded hardware.
  Developed an **automated benchmarking pipeline** using the EuRoC MAV dataset to evaluate trajectory accuracy, tracking performance, and runtime metrics while investigating compiler-level and algorithmic optimizations for improved real-time performance.

- **High Precision Coordinate Measurement System for a Mesh Manufacturing Industry based in USA**
  Developed a computer vision-based solution for defect detection and dimensional measurement in metal mesh manufacturing using **DIPlib** for advanced image processing and **sub-pixel measurement**. The system achieves an accuracy of **300 microns**, identifying irregularities in mesh geometry and dimensions through automated image analysis, thereby strengthening quality assurance and significantly reducing manual inspection time.

- **Defect Detection System for a Textile Industry based in Surat**
  Developed a complete computer vision algorithm for automated defect detection and color classification in metallic Zari thread manufacturing. The system accurately identifies surface defects and color inconsistencies in metallic colors through image processing techniques, enhancing quality control efficiency and minimizing manual inspection errors across production batches.

- **Myoelectric Bionic Arm Prosthesis for EMG-Based Gesture Recognition**
  Developed a myoelectric bionic arm leveraging **EMG signal processing** for **seven-class hand gesture recognition** from forearm muscle activity. Designed a complete pipeline comprising **signal preprocessing, feature extraction, and Machine Learning based classification**, achieving **97% gesture recognition accuracy**. Implemented a **3D-designed, servo-actuated prosthetic prototype** to validate real-time gesture-to-motion mapping and system feasibility.

TECHNICAL SKILLS

- **Programming & Scientific Computing**
  C, C++, Python, MATLAB | Python Libraries: OpenCV, DIPlib, NumPy, Pandas, Matplotlib, scikit-learn, Joblib
- **Computer Vision & Image Processing**
  Visual SLAM (ORB-SLAM3), Stereo Vision, Stereo-Inertial Visual Odometry, Camera Calibration & Rectification, Feature Detection & Matching (ORB), Image Stitching & Panorama Generation, Loop Closure, Pose Estimation
  Image Filtering, Edge Detection, Segmentation, Morphological Operations | Object Detection and Geometric Feature Analysis | Sub-pixel Coordinate Measurement and Dimensional Analysis | Photoclinometry / Shape-from-Shading (Lambertian and Hapke Models) | 2D Data Interpretation for Surface and Depth Estimation
- **3D Modeling & Design**
  Fusion 360, AutoCAD, Autodesk Maya
- **Machine Learning and Deep Learning in Computer Vision**
- **Embedded Systems & IoT**
  GPU: Jetson Orin, Microcontrollers: STM32, ESP32, NodeMCU, Raspberry Pi | Communication Protocols: Node-Red | MQTT, UART, I2C, SPI | IoT System Architectures
- **Deployment, Web & Development Tools**
  Web Technologies: HTML, CSS | Backend & APIs: Flask | Model Serialization & Inference: Joblib | Version Control & Collaboration: Git

SOFT SKILLS

- **Languages-** English (Professional), Hindi (Professional), Marathi (Native),
- **Effective communication**, **adaptability**, and **time management**

Personal Details

Gender- Female | **Date of Birth** - 22 Jan 2005 | **Emails** - Janhavi.pohnerkar@gmail.com, 202301090067@mitaoe.ac.in